Momentum in two dimensions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Conserve momentum separately along perpendicular axes.

SUCCESS CRITERIA - I can...

1. Choose and label two axes.
2. Add momentum components along matching directions.
3. Connect component sums to the final motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two carts approach: one has momentum **5 kg·m/s** *east* and the other **3 kg·m/s** *west*. **Explain why adding their magnitudes** gives the wrong total; **find the signed total**.

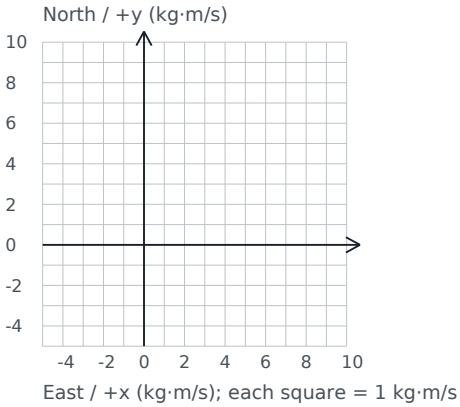
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p_{x,i} = p_{x,f}$; $p_{y,i} = p_{y,f}$; $p_x = mv_x$; $p_y = mv_y$. Use total system momentum in each direction; negligible external impulse. CPA components are supplied.

Worked reasoning: Two objects have momenta (**3**, **0**) and (**0**, **4**) kg·m/s. Their total is (**3**, **4**), magnitude **5 kg·m/s**. **Add components before finding the** magnitude.

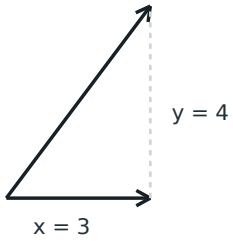
Your turn: Draw a tip-to -tail momentum diagram and **label each axis** with units.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two incoming arrows point *east* and *north*.
1. **Sketch the direction of their** vector sum.
2. Can their ordinary lengths be added for the diagonal magnitude?

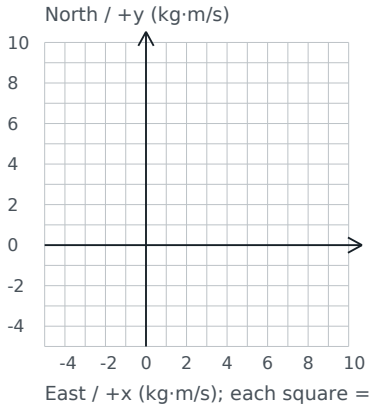


The arrows represent horizontal and vertical components; use their directions to describe the sum.

Marzano check - circle: ① ② ③

04 Apply the physics

Incoming total momentum is **6 kg·m/s** *east* and **8 kg·m/s** *north*. Two objects stick with total mass **2 kg**. Divide each component by **2** to **find velocity components**.



Marzano check - circle: ① ② ③

Momentum in two dimensions

antocciphysics

Continue down each column.

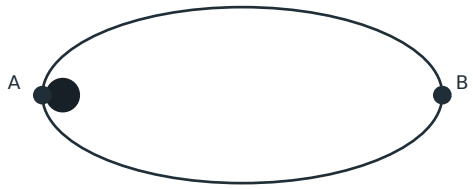
05 Explain your reasoning

Why must both momentum component sums use the same system and same time interval?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

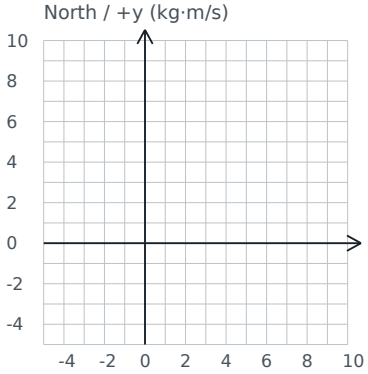
An ideal asteroid orbit has nearest distance **4 AU** and farthest distance **8 AU**. Where is it fastest? Explain with the equal-area law.
Label the orbital features used in your explanation.



Marzano check - circle: 1 2 3

07 Exit check

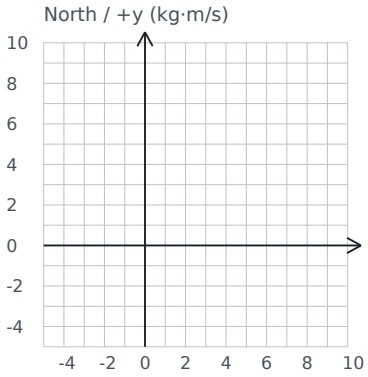
Total momentum is **4 kg·m/s** *west* and **2 kg·m/s** *south*. Combined mass **2 kg**. **Give final velocity components**.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A joined pair of total mass **2 kg** has total momentum **6 kg·m/s** *east* and **8 kg·m/s** *north*. **Find its velocity components** using each momentum component divided by **2**.



Marzano check - circle: 1 2 3